Midterm 2 study guide

How to prepare — Lectures 18–29 and Quizzes 5–9

Last updated: October 19, 2026.

Exam: Midterm 2 · Friday, November 13 · in class. Practice problems: Practice packet. Schedule: Schedule.

Recommended study plan

1. **Scan the learning outcomes below.** Mark each as: confident / shaky / blank.
2. **Go through the lecture notes (18–26) plus Lab 3.** For each outcome you marked shaky or blank, find the matching fill-in, worked example, or Check yourself item. (Skip Veterans Day — no new lecture. Review on Nov 9 is Lecture 26; the exam day page is Lecture 27.)
3. **Build your own "important definitions and formulas" list.** As you read, write down every definition or formula you might need on the exam (matrix sizes, Ax, RREF language, A^k, inverse tests, rank/null/col, determinants).
4. **Rework examples from lecture and quizzes.** Cover the solution, redo Warm-up / Worked / Check yourself / quiz-prep items by hand, then uncover and check using the filled-in notes in Canvas.
5. **Start memorizing your list of definitions and formulas.**
6. **Try the practice midterm problems with no notes.**
7. **Check answers** (end of the practice packet). For every miss, update your definitions/formulas list and redo a similar lecture example.
8. **Memorize the final list** so you can write it from memory before the exam. In class you will not have notes.

Useful Course materials

Source	Role
Lectures 18–29	Core ideas and skills (Unit II)
Quizzes 5–9	Fluency checkpoints (rework those styles; Quiz 5 is Unit I basis review)
Practice packet	Definitions, easy computations, harder items
Boyd & Vandenberghe, *VMLS*	Textbook (free online)
Strang §3.5	Rank, column space, null space (selected portions)

Learning outcomes (Lectures 18–29)

Light review — Quiz 5 / basis (Lecture 14)

1. Explain what a basis accomplishes and what dimension means.
2. Identify a basis and dimension in simple examples.
3. Use span, independence, basis, and dimension accurately when they reappear with matrices (column space, null space, rank).

Lecture 18 — Matrices as data and structured objects

1. Identify matrix dimensions, rows, columns, and entries.
2. Represent datasets, networks, images, or systems as matrices.
3. Perform matrix addition, scaling, and transposition.
4. Recognize zero, identity, diagonal, triangular, symmetric, and sparse matrices.
5. Explain one practical consequence of matrix structure.

Lecture 19 — Matrix-vector multiplication

1. Compute Ax by hand for a small matrix.
2. Interpret Ax as a linear combination of the columns of A.
3. Interpret A as an input-output transformation.
4. Explain how the dimensions of A determine the input and output spaces.

Lecture 20 — Transforming space

1. Predict the effect of scaling, reflection, shear, and rotation matrices.
2. Apply matrices to many 2D or 3D points at once.
3. Plot and compare original and transformed objects.
4. Adapt a short program that implements a transformation.
5. Interpret a matrix as a coordinate or physical-space transformation.

From October 19 through the midterm, class meetings are Lectures 21–31. The bullets below are grouped by those lecture numbers. Nothing from the old lists was removed; systems and elimination share Lecture 21, RREF and one-step dynamics share Lecture 22, and iteration and long-run patterns share Lecture 23.

Lecture 21 — Systems, elimination, and $Ax = b$

1. Translate equations or contextual constraints into $Ax = b$.
2. Interpret $Ax = b$ as asking whether b belongs to the span of the columns of A.